Chapter 27
The Melville Bugt Dyke Swarm of Greenland: A Connection to the 1.5-1.6 Ga Fennoscandian Rapakivi Granite Province?

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Introduction

The continental fit between Laurentia and Fennoscandia during the Precambrian has been hampered by the lack of definitive geological correlations and by paleomagnetic data that are either imprecise or missing at critical times in the geological record. A major time period from about 1.65 to 1.45 Ga is well represented both paleomagnetically and geochronologically in Fennoscandia and particularly by the rapakivi granite province (Rämö and Haapala, 1995). In Laurentia this time period is weakly represented although the 1.59 Ga Western Channel Diabase in NW Canada is an exception (Hamilton and Buchan, 2010; Irving et al., 1972).

Current fits for Laurentia – Fennoscandia employing both geological correlations during the time period both before and after that under consideration suggest that there has been a long-lived connection between the two continental entities between about 1.8 and 1.3 Ga. To date most fits on paleomagnetic grounds are ones in which Laurentia sits in northern latitudes and is rotated clockwise with respect to its present position, so that northern Greenland becomes juxtaposed with northern Norway (e.g. Buchan et al., 2000; Pesonen et al., 2003).

In this paper we present paleomagnetic, U-Pb age and geochemical data from the Melville Bugt (Bay) dyke swarm that trends NNW along the western coast of Greenland (Nielsen, 1990) for about 1,000 km (Fig. 27.1). Prior to our study, only a single paleomagnetic site had been collected from the Melville Bugt dykes (Hamilton et al., 2004) but the paleomagnetic results could not be readily reconciled

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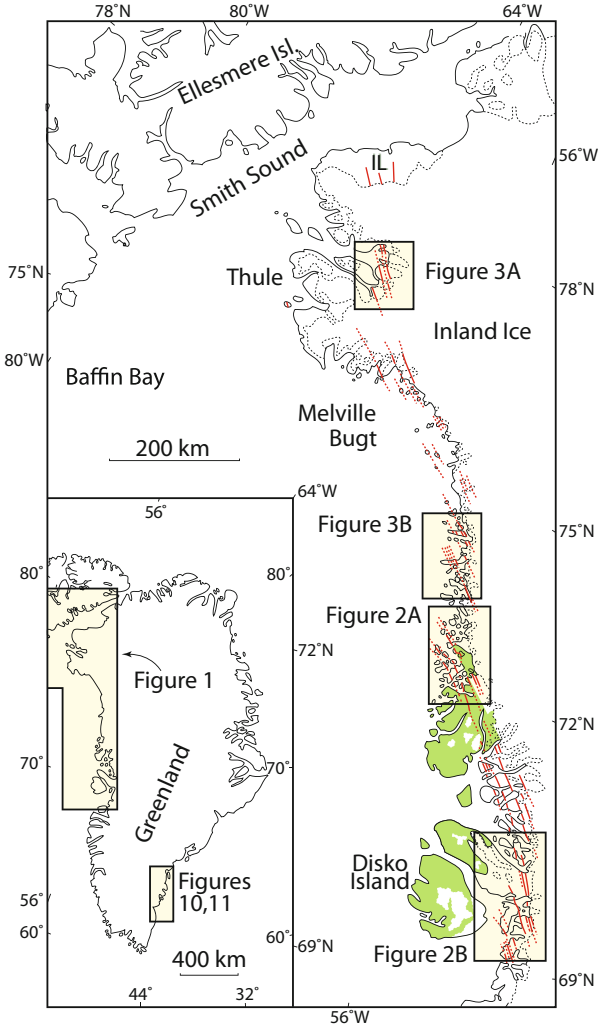


Fig. 27.1 Map showing the full extent of the Melville Bugt dyke swarm. IL: Inglefield Land. Dotted line is the boundary between ice and exposed bedrock; *Green* areas are Tertiary volcanics, uncoloured bedrock areas are Archean and early Proterozoic metamorphic basement rocks. *Boxed areas* are the locations of maps shown in Figs. 27.2 and 27.3. The small outlined area in the map of Greenland shows the location of Figs. 27.10 and 27.11

with Fennoscandian paleomagnetic data. The purpose of this paper is to throw further light upon Fennoscandia-Greenland connections during a critical time period for which paleomagnetic data, particularly in Laurentia, are sparse.

Towards the north the Melville Bugt swarm appears to die out; it is present in Inglefield Land (IL in Fig. 27.1) where dykes are relatively narrow (generally less than 15 m wide, Dawes, 2004) compared to dykes averaging about 100 m across at

the eastern end of Inglefield Bredning (fjord) (northernmost inset map in Fig. 27.1). The swarm cannot be traced into Canada as it passes under Paleozoic strata on Inglefield Land (Dawes, 2004). However if Ellesmere Island has been sinistrally offset during Mesozoic time about 200 km from Greenland along the hypothesized Nares Strait fault (e.g. Denyszyn et al., 2009), then the swarm might be expected to cross Ellesmere Island, but no dykes of the expected trend and width are encountered (Frisch, 1984a, b; Denyszyn, 2008 and helicopter reconnaissance by HCH and SD in 2005 and 2006). To the south the swarm passes under the Greenland ice cap (Fig. 27.1). The western margin of the swarm is exposed but the eastern is not so that neither the true width nor length of the swarm are known. Individual dykes can be up to 100–200 m wide and one extends for at least 400 km and shows a remarkably constant geochemistry over this entire distance (Kalsbeek and Taylor, 1986). The dykes are described by Nielsen (1990) as alkaline trachybasalts.

Field Surveys

In the Summer of 2008 fieldwork by two of us (HCH and MAH) resulted in the collection of 11 paleomagnetic sites from the Melville Bugt swarm (Figs. 27.2 and 27.3), using a portable field drill. A few hand samples were also collected. The collection strategy was to collect most samples from both chilled margins if available because margins, if fresh, tend to preserve primary remanences more so than interiors due to the smaller grain size of remanence carriers (in this case magnetite as deduced from thermomagnetic data by Denyszyn, 2008). Samples were collected from each dyke interior for paleomagnetic comparison but also for U-Pb geochronology and petrography. It has been shown that where the host rock has remained un-melted, chilled margins of a diabase dyke carry a primary geochemical signature which is diagnostic of that dyke and different from other members of the swarm (Halls, 1986; Halls et al., 2008). Therefore major, minor and trace element analyses have been obtained on all chilled margin samples, so that any dykes that have been sampled by more than one paleomagnetic site might be recognized and paleomagnetic statistics adjusted to allow for this duplication.

This collection augments a previous one by HCH and SD in the Thule area of Greenland (Fig. 27.3) where three sites were collected in 2004 and 2005, and a single site collected in the Disko Island area (“Torsukattak”) in the 400 km long dyke at the southern end of the swarm for which a U-Pb age also exists (Fig. 27.2, Hamilton et al., 2004).

Most paleomagnetic sites were collected using both a magnetic and sun compass. The only sites lacking sun orientations, due to cloudy weather, were sites MB6-9, MB 11, 12 and the one north of Disko Island. The presence of sun measurements is extremely important at latitudes of 73–77°N because the steep inclination of the earth’s field necessitates that the rocks be sufficiently non-magnetic that they do not perturb the relatively weak horizontal field that determines the magnetic compass reading.

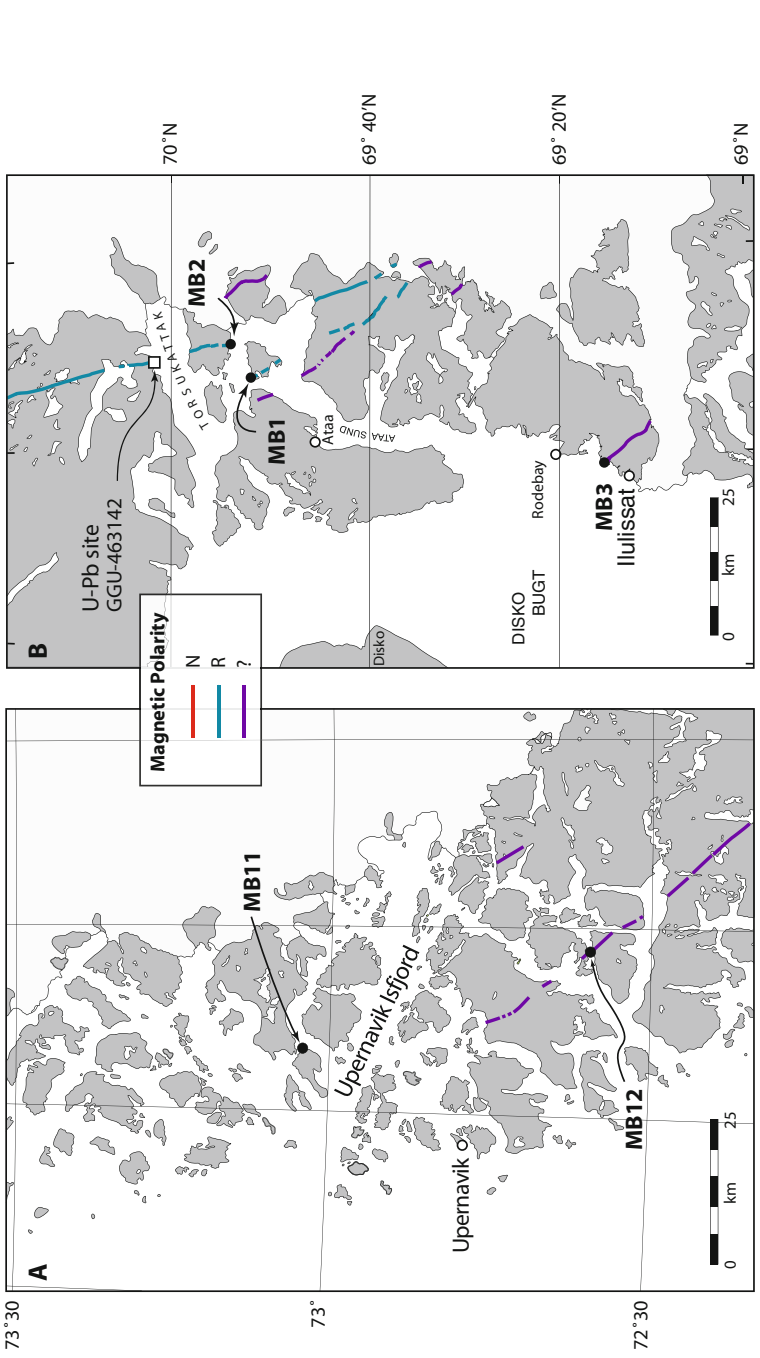


Fig. 27.2 Maps of the southern part of the sampling region showing location of paleomagnetic sites in Melville Bugt dykes: (**a**) Sites MB11 and 12 (**b**) Sites MB1 to 3. *Brown areas* represent-ice-free land; *blue areas* are ice covered

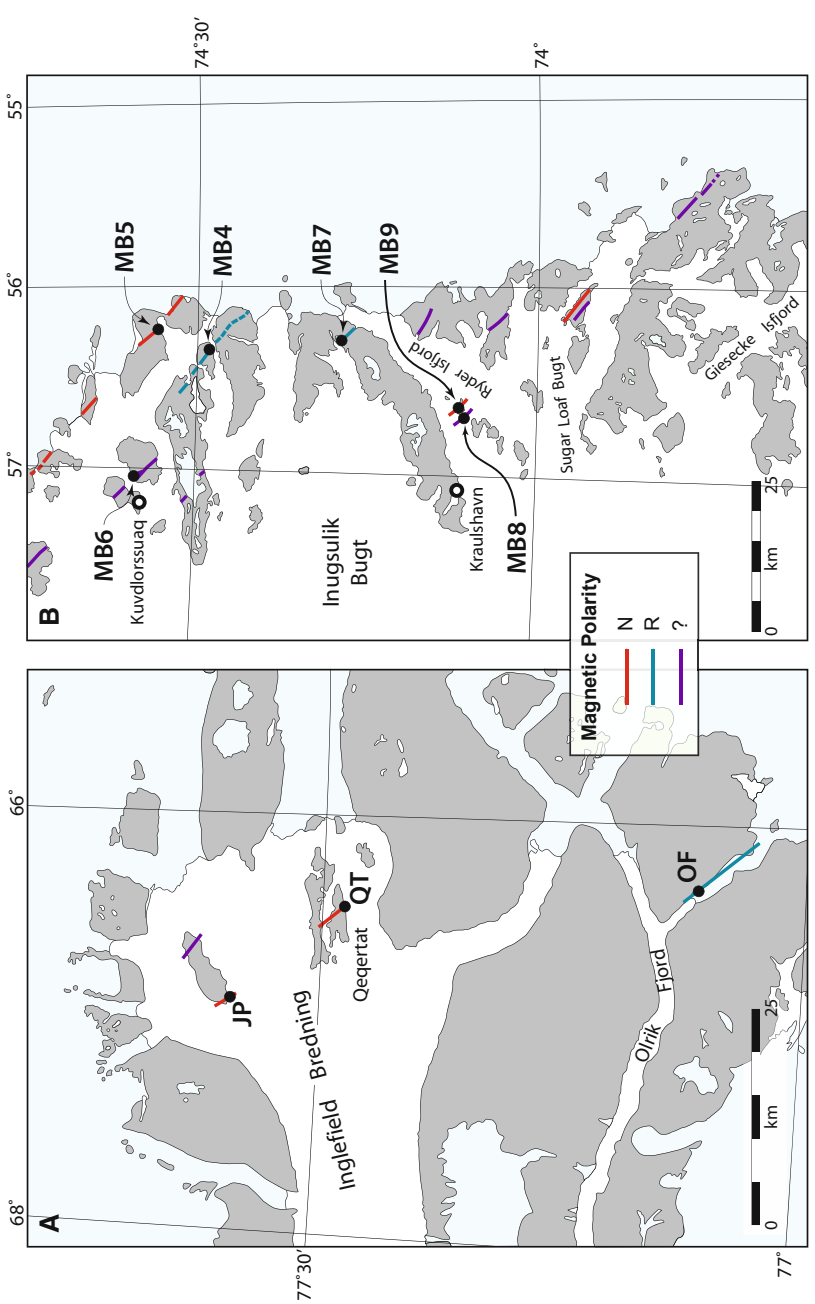


Fig. 27.3 Maps from the northern half of the study area, showing the location of paleomagnetic sites in Melville Bugt dykes. (**a**) Sites OF, QT and JP, (**b**) Sites MB4 to 9. Colours as in Fig. 27.2

Laboratory Methods

Paleomagnetism

Cores or blocks ("samples") were prepared as cylinders, 2.45 × 2.45 cm ("specimens") and at least one specimen per sample was subject to detailed stepwise alternating field demagnetization using a Schonstedt SD-1 instrument and the magnetization measured on a modified DIGICO magnetometer with reproducibility down to magnetization intensities of about 10^{-3} Am^{-1}. At least one specimen per site was thermally demagnetized using a Schonstedt TSD-1 thermal demagnetizer and if the demagnetization proved more successful than AF at separating components, further samples in the site were thermally treated. Samples from one site (MB8) were subject to liquid nitrogen demagnetization (Ozima et al., 1964; Borradaile et al., 2004) in an attempt to remove a strong PEF-like overprint prior to AF treatment. Component separation was achieved by applying Principal Component Analysis to the demagnetization data (Kirschvink, 1980), with acceptance of linear segments having maximum angles of deviation <10° and at least four data points. The AF demagnetization data were subject to an averaging routine (see Halls,1986), once magnetization intensities had fallen to below about 10% of the NRM. This averaging method often extended the AF range of remagnetization circles and/or linear segments, thus improving the definition of magnetization component directions.

U-Pb Geochronology

Most U-Pb analytical work was carried out in the Jack Satterly Geochronology Laboratory (JSGL) at the University of Toronto, with methods following those described in detail by Hamilton and Buchan (2010). Samples for geochronology described here were collected from the freshest, coarsest-grained portions of dykes that ranged in thickness between 100 and 200 m. Generally, only a few hundred grams total of either fresh drill core or hand sample material were crushed and then finely ground using standard jaw crusher and disk/ring mill techniques. The resulting coarse powder was then spoon-fed in a wet slurry slowly over a shaking (Wilfley) table (see Söderlund and Johansson, 2002), left for several minutes, and then a thin band of heavy mineral concentrates were pipetted directly off the table and inspected using a binocular microscope. Using this method, no heavy liquids or magnetic separation techniques were employed. In most cases, hundreds of good quality baddeleyite (ZrO_2) crystals were recovered for U-Pb geochronology from the small sample volumes processed, and showed a serial range in grain size from less than 20 μm to over 200 μm in longest dimension. Highest quality grains of baddeleyite were handpicked under ethanol using a binocular microscope and selected for isotopic analysis. Prior to dissolution, weights of selected fractions were estimated by use of a scaled digital photographic measurement of the length

and breadth of each component grain, and an estimate of the maximum thickness, together with the known density.

Baddeleyite grains were prepared for isotope dilution – thermal ionization mass spectrometry (ID-TIMS) analyses at the JSGL by first washing the fractions in 7 N HNO_3, Milli-Q water and acetone on Parafilm, then loading into a Teflon bomb in a drop of HNO_3. The samples were dissolved in concentrated HF mixed with $^{205}Pb/^{235}U$ spike, and, following conversion to chloride, the samples were loaded, without column chemistry, directly onto Re filaments with Si-gel and phosphoric acid. Isotope ratios were measured on a VG354 mass spectrometer using a Daly detector equipped with digital ion counting. System dead-time corrections during this period were 21 ns for Pb and U. Detector characteristics were monitored during the analytical period using the SRM-982 Pb and CBNM 072/6 U standards. Corrections were 0.05%/amu for Daly mass discrimination and 0.10%/amu for thermal mass discrimination. JSGL procedural blanks are normally at the 0.5 pg and 0.1 pg level for Pb and U, respectively. The measured total common Pb in the samples was generally low and was assigned the common Pb composition of the lab blank (see footnote to Table 27.2).

Sample GGU-463142 was processed and analyzed at the Geological Survey of Canada in Ottawa in 2002. Methods for heavy mineral separation at that time involved crushing and grinding much larger samples (approximately 10 kg), followed by conventional Wilfley table and heavy liquid techniques, and isolation of least paramagnetic heavy minerals using a Frantz isodynamic magnetic separator. Details of this, and the GSC isotope dilution – thermal ionization mass spectrometry (ID-TIMS) methods for U-Pb geochronology, are provided in Hamilton et al. (2001, 2002). Pb and U from the larger fractions prepared for sample GGU-463142 were purified using ion exchange chromatography. Isotopic ratios were measured on a Finnigan-MAT 261 mass spectrometer, using Faraday detectors and a secondary electron multiplier. Lead fractionation was approximately 0.09%/amu. All common Pb was assumed to have the composition of the laboratory blank (see footnote to Table 27.2).

Comprehensive error estimation for each sample in this study was made by propagating all known sources of analytical error, and uncertainties in the composition of the laboratory blanks. Decay constants used are those of Jaffey et al. (1971). Age regressions and errors were calculated, and U-Pb Concordia diagrams were generated using the Microsoft Excel Add-in IsoPlot/Ex v.3.0 (Ludwig, 2003). Concordia ellipse errors and calculated ages are shown at the 2σ (95% confidence) level of uncertainty.

Results

Dyke Petrography

Petrographically all Melville Bugt dykes are virtually identical, being extremely fresh olivine basalt with an equigranular to sub-ophitic texture. They contain about

50% plagioclase with up to about 10% saussuritisation, 25% of a pink clinopyroxene, 10–15% fresh olivine with magnetite-rich cracks, 5% magnetite often associated with a deep brown biotite, and accessory minerals, developed especially where saussuritization is greatest: apatite, quartz, chlorite, epidote. Hornblende as an alteration product of clinopyroxene is largely absent and primary grains are rare. Only samples at site MB1 lack olivine.

Paleomagnetism

The full paleomagnetic results are presented in Table 27.1 and specific aspects of specimen behavior under demagnetization are illustrated in Fig. 27.4. More or less antipodal directions are present (Fig. 27.4a, b, f): the swarm can be divided into four dykes that have a down to the SW characteristic direction and five dykes that have intermediate negative inclinations towards the NNE. Those dykes having the steepest negative inclinations (see Fig. 27.4a, b) generally have the least overprinting by present Earth's field (PEF)-like components. Occasionally a specimen with a strong PEF overprint will attain a stable end-point which is shallower (Fig. 27.4c). Those samples that have a progressively larger PEF component fail to achieve end-points but move towards the NE assuming shallow inclinations of ±5° at the limit of AF demagnetization (e.g. Fig. 27.4d). Thermal demagnetization appears incapable of improving component separation (Fig. 27.4e). One site (MB7) attains stable end points at a shallow inclination for both AF and thermal treatment (Fig. 27.4g, h). Another site (MB8) despite having several beautifully exposed narrow fine-grained satellite dykes has an extremely intense NRM ($\sim 10^2$ Am^{-1}) overprint that was almost vertical and suspected to be a PEF component. Samples from this site were cooled in field-free space down to liquid nitrogen temperatures (120°K) in an effort to remove the PEF component prior to AF treatment (see Borradaile et al., 2004 for an example of applying this method to diabase). The results (Fig. 27.4i, j) show treatments for two specimens from the same sample. In both cases the low temperature treatment removes a significant fraction (>60%) of the vertical overprint, but the remanence remaining has widely different directions. After step-wise AF demagnetization this remanence gradually moves back to the PEF one. The method did not result in the recovery of a coherent component underlying the PEF and therefore was discontinued.

A summary of sample paleomagnetic directions at each site is given in Fig. 27.5. Note that despite most sites moving from a steep down component, only site MB2 is streaked in such a way to suggest the incomplete removal of the steep component and hence an underestimate of the negative inclination. Two further sites (MB1 and 7) also showed an unexplained streaking in the direction of declination, but one that would not affect the inclination.

In terms of paleomagnetic components, the two antipodal ones (A and B) are considered to be primary because each characterises different dykes that are geographically intermixed (Fig. 27.3b). In addition the chilled margins show the least contamination with the PEF and other secondary components. The host rocks,

Table 27.1 Paleomagnetic results

							N	DEC (°)	INC (°)	α_{95} (°)	k	PLAT (°N,–°S)	PLON (°E)	dp (°)	dm (°)	λ (°)	DEC′	INC′	SITE	N	DEC (°)	INC (°)	α_{95} (°)	k
Site	LAT (°N)	LON (°W)	Width (M)	Strike (°)	DIP	NT	Primary component													Secondary component				
MB1	69.8669	50.6201	7.5	345	90	8	8	39.3	–44.5	7	64	10.1	274.1	5.5	8.8	–26	33.1	–40.1	MB1					
MB2	69.8984	50.4479	110	NW	>80°SW	6	4	26.9	–14.9	11.7	62	–10.4	282.5	6.2	12.0	–8	22.2	–7	MB2					
MB3	69.2668	51.0375	50	330	90	7													MB3	6	33.7	47.5	13.5	26
MB4	74.4829	56.3348	60	NW	90	10	5	30.3	–29.3	9.1	72	2.2	274.6	5.5	10.0	–16	31.6	–28.3	MB4					
MB5	74.5579	56.2202	50	308	80°SW	7	6	209.1	46.6	4.9	186	14.2	277.5	4.1	6.3	28	210.3	46	MB5					
MB6	74.5834	57.0260	55	340	90	8				Erratic									MB6	3	204.7	–70.7	15.3	66
MB7	74.2886	56.2729	70	NW	90	9	5	51.9	–8.5	9.6	64	–5.4	251.7	4.9	9.7	–4	53.1	–7.2	MB7					
MB8	74.1068	56.6731	45	315	90	17													MB8	6	154.9	89.2	5.2	166
MB9	73.9129	55.9418	135	320	90	10	6	201.2	50.1	5.6	142	15.7	285.2	5.0	7.5	31	202.1	48.8	MB9					
MB11	73.0373	55.6619	45	325	80°SW	12													MB11	6	226.2	–41.4	17.5	16
MB12	72.5981	55.1217	210	NW	~90	8				Erratic									MB12					
QT	77.4781	66.4256	100	307	80°NE	9	8	195.9	28.8	8.8	41	3.3	278.2	5.3	9.7	15	208.9	30.9	QT					
JP	77.5933	66.8817	70	325	NO CM	9	7	209.1	44.2	3	40	15	266.2	2.4	3.8	26	222.5	44.8	JP					
OF	77.1058	66.2670	200	330	85°SW	9	7	13.1	–6.7	10.5	25	–9.2	280.5	5.3	10.5	–3			OF					
OF	Margin					9	5	19.8	–23.2	7.1	118	–0.2	274.4	4.0	7.6	–12	31.5	–24.2	OF					
Mean pole position							9			8.7		5.0°N	273.8°E				33.1	–31.2	Mean					

Notes: North poles are given for SW-directed remanences but South poles for NE-directed ones. DEC′, INC′: Declination, Inclination for common site locality 75°N, 55°W. NT/N are number of samples collected per site/providing data. For the pole position, N is the number of sites and the confidence limit is A_{95}.

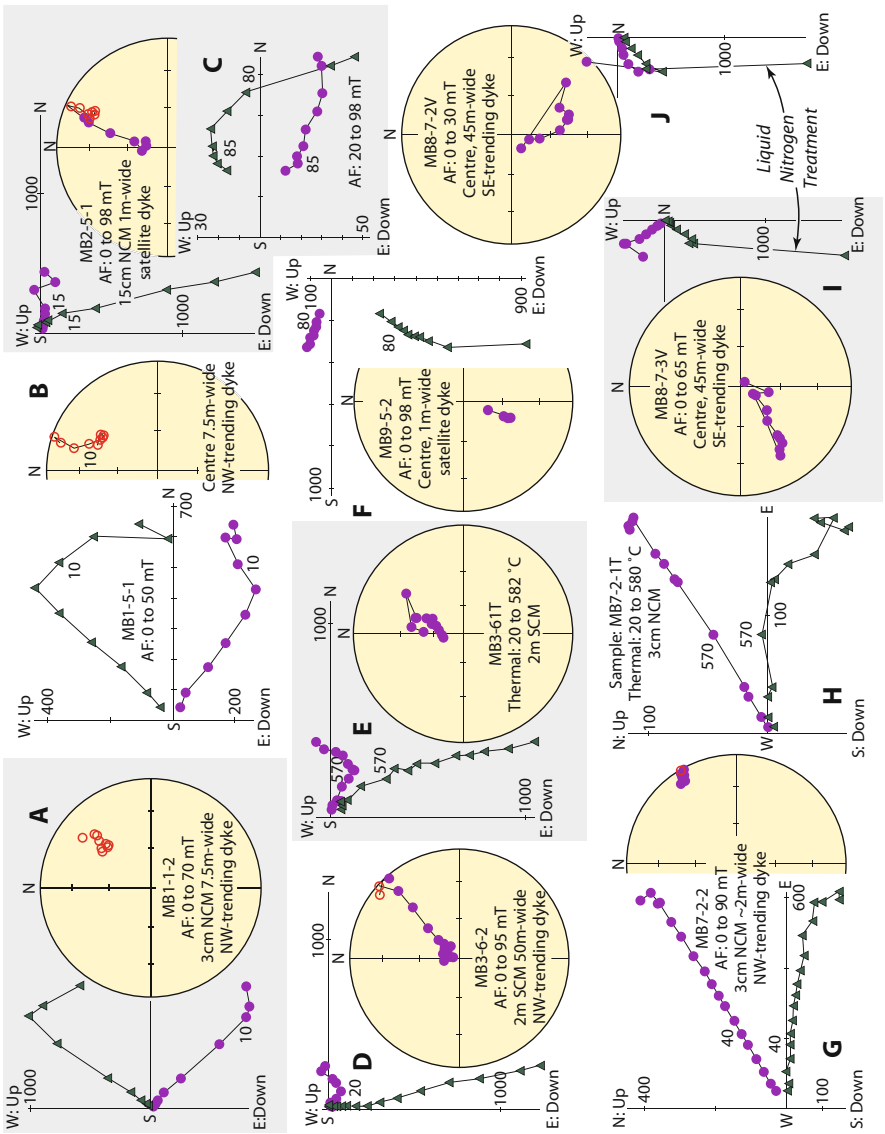


Fig. 27.4 (continued)

coarse-grained granitic gneisses in amphibolite facies, were not sampled (previous experience with such rocks leads to an expectation of weak and unstable remanences) so a baked contact test, needed to conclusively prove a primary origin for the remanence, was not attempted. However, the ca. 720 Ma Franklin dyke swarm is in close proximity to the Melville Bugt swarm in the Inglefield Bredning area and its paleomagnetic direction, shown to be primary by a positive baked contact test (Denyszyn et al., 2009) is distinct from that of the Melville Bugt dykes. This observation rules out any regional overprinting of Melville Bugt dykes after 720 Ma. Apart from the PEF-like remanence, defined at site MB8, the only other components found in Melville Bugt dykes were one of unknown origin (34°/48°, N = 6, $\alpha_{95} = 14°$) found at site MB3, and a steep negative one to the SW (205°/–71°, N = 3, $\alpha_{95} = 15°$) from site MB6 which may represent an overprint from the nearby Tertiary volcanic province (see Fig. 27.1 for the distribution of flood basalts and picrites) which gives a mean paleomagnetic direction of 163°/–70°, N = 44, $\alpha_{95} = 4°$ with some sites having a SW declination (Riisager et al., 2003). Therefore despite its proximity and the thickness of the volcanic sequence (5–6 km) the Tertiary volcanic province has had a minimal effect upon the magnetism of the Melville Bugt dykes.

Equal area stereonets of the paleomagnetic data for each Melville Bugt site are given in Fig. 27.5, and in Fig. 27.6 as a stereonet summary of the final Melville Bugt data, using site directions with one polarity reversed (Fig. 27.6a), with polarities separated (Fig. 27.6b), and with directions expressed as virtual geomagnetic poles (Fig. 27.6c).

Fig. 27.4 (continued) Equal area and vector diagrams showing the response of selected Melville Bugt specimens to AF and thermal demagnetization. On stereoplots *solid/open symbols* are downward/upward-pointing magnetizations. On vector diagrams *circles* represent the projection of the tip of the remanence vector, after each AF or thermal demagnetization step, onto the *horizontal plane* and *triangles* represent the projection onto a vertical W-E plane, except for diagrams g and h where the projection plane is N-S. SCM and NCM are, respectively, the southwestern and northeastern dyke margins (e.g. "3 cm NCM. . ." implies the sample was collected 3 cm inside the northeastern chilled margin). Diagrams (**a**) and (**b**): two samples from the same site that yield a well-defined stable end-point corresponding to an interpreted primary component. (**c**), (**d**) and (**e**) are examples of the response under AF and thermal demagnetization of samples that are strongly overprinted by a PEF-like component but which move to shallow inclinations with NE declination without attaining a stable end-point. (**f**) is an example of a specimen that attains a stable end point corresponding to a suspected primary component that is antipodal to that in (**a**) and (**b**). (**g**) and (**h**) are examples of AF and thermal demagnetization on the same sample which shows that in some sites like MB7 the shallow NE component appears to be real. (**i**) and (**j**) represent an attempt, using liquid nitrogen cooling in field-free space, to eliminate a strong vertical component at site MB8, in an attempt to recover an underlying older component. Note that while the specimens were strongly demagnetized by the technique, the residual remanence direction was quite different, and moreover that subsequent AF demagnetization merely returned the specimens to their initial NRM direction

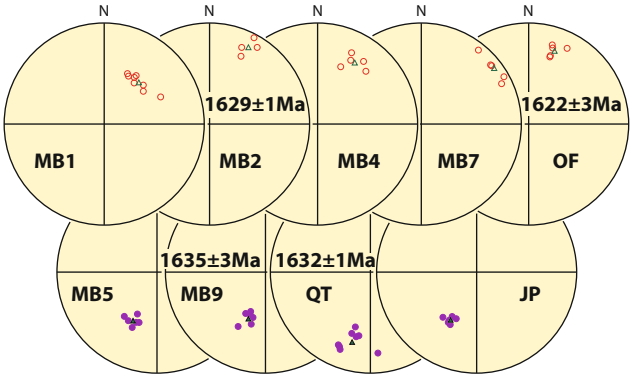


Fig. 27.5 Summary equal area stereonets showing sample directions for each of the Melville Bugt sites that are interpreted to be carrying a primary remanence. Note that only MB2 shows streaking in the expected direction between the primary and the PEF-like component. Sites MB1 and MB7 show streaking but not in a way that could account for error in the inclination of the primary direction due to unresolved components

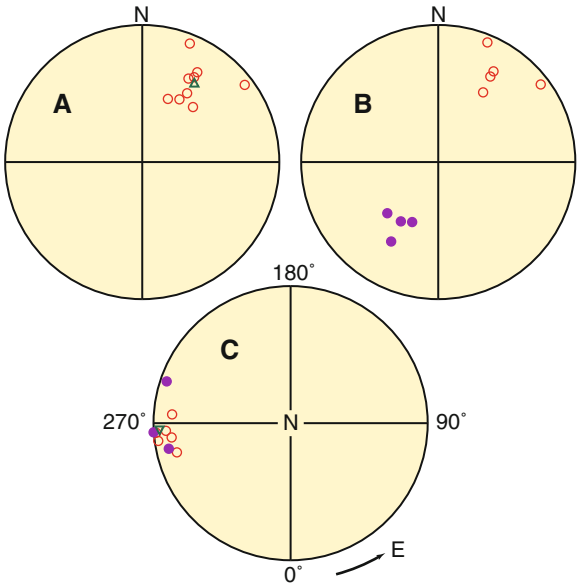


Fig. 27.6 Summaries of site mean data for Melville Bugt dykes: (**a**) Site directions with antipodal directions reversed; (**b**) Site directions without reversing data. *Solid/open* symbols are downward/upward pointing magnetizations; directions corrected to a common site at 77°N, 55°E, the *triangle* in A is the mean direction. (**c**) Virtual geomagnetic poles for each site, showing them to be clustered about the equator. View of earth is looking down from the N pole, with longitudes marked in degrees east. *Solid/open points* are poles in the southern/northern hemisphere. *Triangle* is the mean pole position

U-Pb Geochronology

We present the results of U-Pb isotopic dating of four Melville Bugt diabase dykes in Table 27.2, with graphic representations of the data in Concordia diagrams in Fig. 27.7(a–d). These four individual dykes span almost the entire known width of the swarm as recognized in west and northwest Greenland, as well as the north-south distribution from Inglefield Bredning to Ilulissat (a distance of roughly 1,000 km). Two of these are from dykes that yielded the down-to-the-SW remanence (MB9, QT), and two are from dykes that gave the up-to-the-NNE remanence (MB2, OF), both of which are interpreted to be primary.

Baddeleyite was successfully recovered from all four dykes. Invariably, these consisted of striated, pale to medium (and rarely dark) brown euhedral blades and blade fragments ranging up to ~150 μm in the longest dimension (but typically <60 μm). Fractions chosen for analysis normally comprised between 2 and 20 grains, but single grain fractions were also analyzed (Table 27.2). Uranium concentrations in the baddeleyite crystals are relatively high, mostly between 400 and 1,000 ppm, but reach up to ~2,200 ppm U. Estimated Th/U ratios for all fractions are low (<0.06) and typical of baddeleyite crystallizing in mafic or gabbroic magmas (Heaman and LeCheminant, 1993). Data from most of the samples plot close to the Concordia curve (all <2.5% discordant; Table 27.2, Fig. 27.7).

Site MB9 (sample HMB08-013): This NW-trending (320°) dyke reached up to 135 m in thickness where it was sampled on a small island (Paornarqutût) inside Ryder Isfjord, east of Kraulshavn – in the approximate geographic centre of the known Melville Bugt swarm (Figs. 27.1 and 27.3b). The middle of the dyke at this site was characterized by roughly 50 cm-diameter ovoidal patches of coarse to pegmatitic, locally ophitic gabbro rich in ilmenite and apatite. Baddeleyite was abundant in these segregations, which are interpreted to represent pockets of evolved, trapped melt. Zircon was not observed.

Data for four fractions, comprising between 4 and 6 grains of baddeleyite each, are variably discordant and regress to yield an upper intercept age of 1,635.0 ± 2.7 Ma (2σ, 76% probability of fit; Fig. 27.7a). This result is interpreted to represent the age of emplacement and crystallization of the Melville Bugt dyke at this site, and constrains the remanent magnetization direction here – SW declination and down ("normal" polarity) – to have been characteristic at this time. Although poorly constrained, the lower intercept of the U-Pb array, at ~215 Ma, may reflect disturbance during Mesozoic opening of the Labrador Sea.

Site QT (sample QT-1): The central portion of Qeqertat Island, at the head of Inglefield Bredning in northwestern Greenland, is cut by a conspicuous NW-trending (307°) dyke with a thickness reaching approximately 100 m (Fig. 27.3a). This region encompasses the most northern and eastern limits of demonstrable Melville Bugt dykes. The Qeqertat dyke may continue along strike as the same dyke sampled at site JP as both also exhibit normal magnetic polarity. However, their paleomagnetic directions are distinct at the 95% confidence level and the geochemistry of their chilled margins is dissimilar, suggesting two separate intrusions.

Table 27.2 U-Pb analytical results for baddeleyite from Melville Bugt diabase dykes, west Greenland

Sample fraction	Analysis no.	Description	Weight (μg)	U (ppm)	Th/U	Pb* (pg)	Pb_C (pg)	$^{206}Pb/^{204}Pb$	$^{206}Pb/^{238}U$	±2σ	$^{207}Pb/^{235}U$	±2σ	$^{207}Pb/^{206}Pb$	±2σ	$^{206}Pb/^{238}U$ Age (Ma)	±2σ	$^{207}Pb/^{235}U$ Age (Ma)	±2σ	$^{207}Pb/^{206}Pb$ Age (Ma)	±2σ	Disc. (%)	Corr. Coeff.
HMB08-013 (site MB9) – Ryder Isfjord; 320° strike																						
Bd-1	MAH8163	4 mbr long prismatic	0.2	428	0.009	23	0.5	3,189	0.28533	0.00063	3.9557	0.0128	0.10055	0.00019	1,618.1	3.1	1,625.2	2.6	1,634.3	3.6	1.1	0.8173
Bd-2	MAH8164	5 mbr short blades & frags	0.2	495	0.015	27	0.4	4,530	0.28593	0.00061	3.9613	0.0115	0.10048	0.00015	1,621.2	3.1	1,626.3	2.3	1,633.0	2.8	0.8	0.8617
Bd-3	MAH8165	6 mbr blades & frags	0.2	1,268	0.010	69	0.4	12,194	0.28611	0.00053	3.9661	0.0093	0.10054	0.00009	1,622.1	2.7	1,627.3	1.9	1,634.0	1.7	0.8	0.9273
Bd-4	MAH9012	6 mbr blades & frags	0.2	1,039	0.007	55	0.9	4,205	0.28092	0.00056	3.8888	0.0110	0.10040	0.00016	1,596.0	2.8	1,611.4	2.3	1,631.5	2.9	2.5	0.8443
QT-1 (site QT) - Qeqertat Island, inner Inglefield Fjord; 307° strike																						
Bd-1	MAH0100na	2 mbr long prismatic	0.5	537	0.040	54	0.2	5,332	0.28150	0.00068	3.8994	0.0118	0.10047	0.00014	1,598.9	3.4	1,613.6	2.5	1,632.7	2.5	2.3	0.9011
Bd-2	MAH0101	2 pbr thin blades	0.5	383	0.021	52	0.3	11,943	0.28324	0.00055	3.9220	0.0094	0.10043	0.00009	1,607.6	2.8	1,618.2	1.9	1,632.0	1.8	1.7	0.9270
Bd-3	MAH0102na	3 mbr blades & frags	0.5	601	0.047	82	0.3	15,743	0.28413	0.00056	3.9334	0.0092	0.10040	0.00010	1,612.1	2.8	1,620.6	1.9	1,631.6	1.8	1.3	0.9105
GGU-463142[a] (same dyke as site MB2) – Torsukattak; 356° strike																						
Bd-A	Z7428-A	14 m-dbr, large blades	4.2	1,049	0.037	1200	10.0	7808	0.28524	0.00056	3.9431	0.0097	0.10026	0.00010	1,617.7	2.8	1,622.6	2.0	1,628.9	1.8	0.8	0.9286
Bd-B	Z7428-B	5 m-dbr, blades and frags	4.2	970	0.028	1110	4.6	15,850	0.28608	0.00056	3.9561	0.0097	0.10030	0.00009	1,621.9	2.8	1,625.3	2.0	1,629.6	1.7	0.5	0.9404
Bd-C	Z7428-C	13 mbr, large blades	8.8	607	0.036	1460	10.5	9,028	0.28560	0.00058	3.9522	0.0100	0.10036	0.00009	1,619.5	2.9	1,624.4	2.0	1,630.8	1.7	0.8	0.9420

Table 27.2 (continued)

Sample fraction	Analysis no.	Description	Weight (μg)	U (ppm)	Th/U	Pb* (pg)	Pb_C (pg)	$^{206}Pb/^{204}Pb$	$^{206}Pb/^{238}U$	±2σ	$^{207}Pb/^{235}U$	±2σ	$^{207}Pb/^{206}Pb$	±2σ	$^{206}Pb/^{238}U$ Age (Ma)	±2σ	$^{207}Pb/^{235}U$ Age (Ma)	±2σ	$^{207}Pb/^{206}Pb$ Age (Ma)	±2σ	Disc. (%)	Corr. Coeff.
Bd-D	Z7428-D	20 mbr, large blades	5.6	678	0.037	1040	5.0	13,650	0.28580	0.00059	3.9495	0.0100	0.10023	0.00009	1,620.5	3.0	1,623.9	2.1	1,628.3	1.7	0.5	0.9400
OF-2 (site OF) - Olrik Fjord; 330° strike																						
Bd-1	DWD4671	1 mbr short prismatic	0.6	975	0.060	156	4.8	2,247	0.28765	0.00065	3.9625	0.0158	0.09991	0.00028	1,629.8	3.3	1,626.5	3.2	1,622.4	5.2	–0.5	0.7356
Bd-2	DWD4672	1 mbr short prismatic	0.9	406	0.039	98	2.0	3,453	0.28683	0.00081	3.9493	0.0156	0.09986	0.00024	1,625.7	4.1	1,623.8	3.2	1,621.5	4.4	–0.3	0.8036
Bd-3	DWD4673	1 mbr long prismatic blade	0.6	2,173	0.011	351	1.9	12,609	0.28519	0.00050	3.9379	0.0361	0.10014	0.00089	1,617.5	2.5	1,621.5	7.4	1,626.8	16.5	0.6	0.2393

Notes: All analyzed fractions represent hand-picked, highest quality baddeleyite crystals free of inclusions or adhered/coating phases. Abbreviations: pbr – pale brown; mbr – medium brown; dbr – dark brown; fl – flat; frag - fragment. Pb* is total amount (in picograms) of radiogenic Pb. Pb_C is total measured common Pb (in picograms) assuming the isotopic composition of laboratory blank: ROM [GSC]- 206/204 – 18.221 [18.2]; 207/204 – 15.612 [15.37]; 208/204 – 39.360 [37.4] (errors of 2%). Pb/U atomic ratios are corrected for spike, fractionation, blank, and, where necessary, initial common Pb; 206Pb/204Pb is corrected for spike and fractionation. Th/U is model value calculated from radiogenic 208Pb/206Pb ratio and 207Pb/206Pb age assuming concordance. Disc. (%) – per cent discordance for the given 207Pb/206Pb age. [a]Analyzed at the GSC Geochronology Lab, Ottawa, Canada.

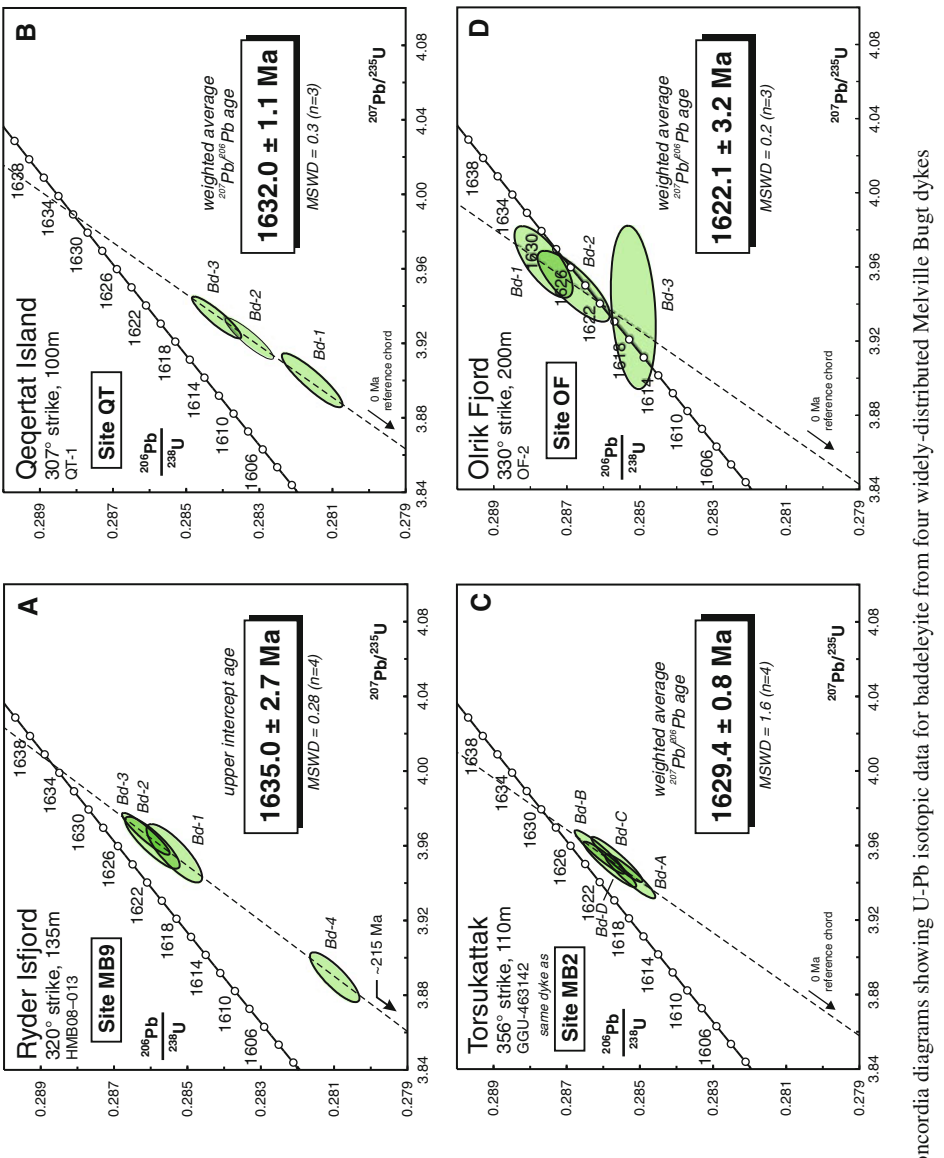


Fig. 27.7 Concordia diagrams showing U-Pb isotopic data for baddeleyite from four widely-distributed Melville Bugt dykes

Three fractions of baddeleyite from sample QT-1 are 1.3–2.3% discordant but are collinear and yield uniform $^{207}Pb/^{206}Pb$ ages – 1,631.6–1,632.7 Ma (Table 27.1; Fig. 27.7b). A weighted average $^{207}Pb/^{206}Pb$ age for all three fractions yields 1,632.0 ± 1.1 Ma (2σ, 74% probability of fit), which we interpret to represent the intrusion and crystallization age of the Melville Bugt diabase dyke at this locality. Like the 1,635 Ma MB9 dyke, the QT dyke yields a magnetic remanence direction, interpreted to be primary, that is oriented to the SW and down (Fig. 27.5), suggesting that this polarity persisted from at least 1,635–1,632 Ma.

Site MB2 (sample GGU-463142): Due east of the large island of Disko, and approximately 80 km north of Ilulissat, the southern end of the large, semi-continuous 400 km-long dyke is exposed on the shores of the fjord Torsukattak (Fig. 27.2b). Further south, its likely extension is covered by the Inland Ice. Sample GGU-463142 was collected from this dyke on the north shore of Torsukattak in 2001 by GEUS (Denmark and Greenland Geological Survey) geologists. Here, the dyke is approximately 70 m thick and strikes almost N–S (354°). Initial dating and paleomagnetic results from this sampling were presented by Hamilton et al. (2004). This dyke undoubtedly continues south along strike across the fjord where it was sampled for paleomagnetic study by us (HCH and MAH) in 2008 (Fig. 27.2b). At site MB2, the dyke also strikes NNW and widens to roughly 100 m in thickness.

U-Pb data for four multigrain fractions of baddeleyite (5–20 grains each) from this sample are tightly clustered and only 0.5–0.8% discordant (Fig. 27.7c). $^{207}Pb/^{206}Pb$ ages are very uniform, ranging narrowly between 1,628.3 and 1,630.8 Ma (Table 27.2). A simple weighted average $^{207}Pb/^{206}Pb$ age for all four fractions is 1,629.4 ± 0.8 Ma (2σ). We interpret this result to represent a rigorous estimate of the age of emplacement of the large Melville Bugt dyke at this locality. Because the MB2 site paleomagnetic data records a remanence magnetization with a NE (and up) declination, by extension we also interpret this age to constrain the timing of a magnetic reversal between that recorded in the QT dyke (1,632 ± 1 Ma) and the MB2 dyke (1,629 ± 1 Ma).

Site OF (sample OF-2): At the head of the long E–W Olrik Fjord in the Thule-Inglefield Bredning region, a wide Melville Bugt dyke (∼200 m thick) parallels the eastern wall of a NW-trending glacier-fed valley (Figs. 27.1 and 27.3a). Here the dyke strikes 330°, and a representative coarse-grained piece from paleomagnetic site OF was collected for geochronology (Denyszyn, 2008). Abundant baddeleyite was recovered from sample OF-2.

Three single grains of fresh, tan-coloured, striated and prismatic baddeleyite from this sample were analyzed individually (Table 27.2). Although the results generally straddle concordia (0.6 to –0.5% discordant; Fig. 27.7d), the sample:blank ratios for these analyses are slightly higher than in the other samples and thus individual analytical uncertainties are also greater. $^{207}Pb/^{206}Pb$ ages for these fractions range from 1,621.5 to 1,626.5 Ma. Nonetheless, a relatively precise weighted average $^{207}Pb/^{206}Pb$ age, controlled by Bd-1 and Bd-2, yields a date for the timing of emplacement and crystallization of this large dyke at 1,622.1 ± 3.2 Ma (2σ, 82% probability of fit). Melville Bugt diabase from site OF also carries the upward,

NE-directed remanence, consistent with the result from the slightly older dyke at MB2 (1,629 ± 1 Ma).

In summary, the precise U-Pb geochronological results presented here demonstrate that the timing of emplacement of the widespread Melville Bugt dyke swarm lasted from at least 1,635 to 1,622 Ma, spanning some 13 Myr. Detailed results from within the geographic distribution of the swarm (as currently recognized) reveals no obvious pattern of intrusion. Dykes appear to be intermingled in terms of age and magnetic polarity, though the trends are all subparallel, and we have seen no evidence of cross-cutting relationships within the swarm. The geochronology results (Table 27.2 and Fig. 27.7) show conclusively that the remanent magnetization with SW declination, as exemplified by the dated sites MB9 and QT, is older than the NE-directed remanence found at dated sites MB2 and OF. Because of the small temporal uncertainties on the nearest dykes (age-wise) recording a reversal in remanence direction (i.e. QT & MB2), this polarity switch must have happened within a narrow window between 1 and 5 Myr, centred on ca. 1,630 Ma.

Interpretation

An alternative model to the conventional paleomagnetic one of placing Laurentia in the northern hemisphere for the time period 1,450–1,600 Ma (Buchan et al., 2000, Pesonen et al., 2003, Lubnina et al., 2010) is to reverse the poles and place it in the southern one. The effect of this is to return Laurentia to a position not unlike that existing today with the tip of Greenland pointing southwards, rather than northwards in present day co-ordinates. In Figs. 27.8 and 27.9 configurations for Laurentia versus Fennoscandia are shown, one at ~1.5 Ga and one at ~1.6 Ga respectively. What becomes apparent is that the Melville Bugt dyke swarm with some adjustments of

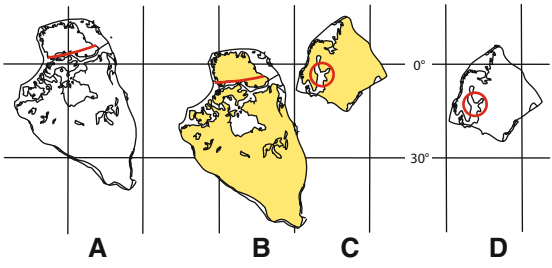


Fig. 27.8 Positions of Laurentia and Fennoscandia for the time ~1.5 Ga based on the average mean poles given by Salminen and Pesonen (2007), together with perturbations in the position of the mean poles allowed by their uncertainty that permit a direct linkage of the Melville Bugt swarm (*red line*) with the Fennoscandian rapakivi province (*red circle*). (**a**) Mean pole for Laurentia (Salminen and Pesonen 2007); (**b**) Mean pole for Laurentia with a displacement allowed by the A95 value (13°); (**c**) mean pole for Fennoscandia with displacement allowed by A95 value (11°), (**d**) mean pole for Fennoscandia (Salminen and Pesonen, 2007)

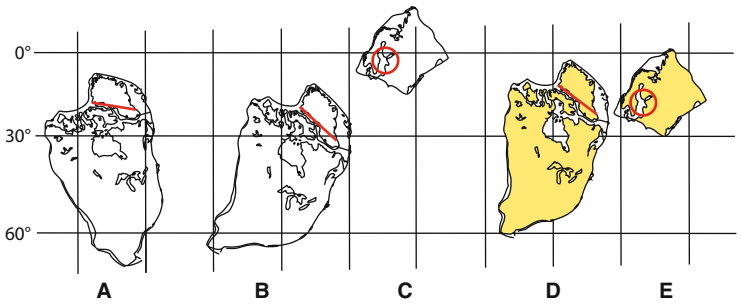


Fig. 27.9 Similar to Fig. 27.8, except that the time period is for ~1.6 Ga. (**a**) position of Laurentia using paleomagnetic data from the 1.59 Ga Western Channel Diabase (Irving et al., 1972; Hamilton and Buchan, 2010); (**b**) the position of the Melville Bugt dykes based upon the pole data in Table 27.1 (**c**) the position of Fennoscandia based on the pole for the Sipoo dykes (Mertanen and Pesonen,1995) and shows that correlation of the ~1630 Ma Melville Bugt dykes (*red line*) with the Fennoscandian rapakivi province (*red circle*) is poor. (**d**) location of Laurentia based on an adjustment of the Melville Bugt pole position that lies within its 95% confidence circle. (**e**) an adjusted position of Fennoscandia based on the premise that the magnetization of the 1,633 Ma Sipoo dykes has an inclination of –30° (see text). A comparison of figures (**d**) and (**e**) shows that the correlation between Melville Bugt dykes and Fennoscandian rapakivi province is now improved but not as close if only the data from the Western Channel Diabase are used to define the position of Laurentia

the paleomagnetic data (see below) approximately points towards the Scandinavian rapakivi granite province, and furthermore that the configuration of Greenland and Fennoscandia may have remained the same throughout the lifetime of rapakivi magmatism. The Melville Bugt dyke swarm has been an enigma because it occurs in isolation with no obvious connection to other coeval magmatic provinces nor to plume centres, a common feature of other major dyke swarms.

If the paleomagnetic configurations shown in Figs. 27.8 and 27.9 are allowable, it means that the source for the Melville Bugt swarm may be represented by the Fennoscandian 1.5–1.6 Ga rapakivi granite province. In this province diabase dykes are associated with rapakivi plutons and generally trend away from them. Cross-cutting relations suggest an order of emplacement, from older to younger, of diabase dykes, rapakivi granite and finally quartz porphyry dykes. It is generally accepted that the rapakivi plutons represent melting of the lower crust, perhaps by a local mantle-derived basaltic melt (e.g. Rämö and Haapala, 1995). The availability of basaltic magma associated with these intrusions is exemplified by the Breven-Hällefors dyke swarm in Sweden. This swarm is composed of over 850 dykes and is at least 150 km wide (Fig. 27.1 of Risku-Norja, 1992). Within the swarm are at least four major dykes several 100 m thick, one of which is about 1 km wide and on the basis of aeromagnetic data, at least 100 km long. The western extent of the swarm is obscured beneath Paleozoic strata and ultimately by the Protogine zone. However the enormous size of these master dykes indicates the capability of rapakivi – associated magmatism to rapidly produce copious amounts of basaltic magma as dykes. In many cases magma-mixing observations suggest the basaltic magma to have been

coeval with felsic magma. The Breven-Hällefors dyke swarm lies on strike with a major rapakivi pluton in Estonia. Other plutons (Wiborg, Åland) are also associated with dykes of more modest proportions. A system of diabase dykes (Sipoo and Häme swarms) lie adjacent to the Wiborg granite. The total age range from U-Pb dating for the Fennoscandian rapakivi province is about 150 Myr (from ~1,650 to 1,500 Ma) so that the Melville Bugt dykes would be associated with the early stages of the rapakivi magmatic event.

Paleomagnetically the Häme and Breven-Hällefors swarms are strongly overprinted by a PEF component (Neuvonen, 1967; Bylund, 1985) although some diabase dykes from the latter study and quartz-porphyry dykes associated with the Wiborg granite give a streaked distribution from a PEF direction to a shallow up, N to NE component (Neuvonen, 1986; Mertanen and Pesonen, 1995). One important observation is that for the Sipoo swarm, the diabase dykes record a remanence which is both older and of opposite polarity to the NE component recorded by the quartz-porphyry dykes. The evidence is that the freshest dyke, site SB, which is farthest from the Wiborg granite, records the SW component, whereas in another more altered dyke, closer to the granite, both the SW *and* NE components are present, suggesting that the alteration was produced during the intrusion of the younger Wiborg granite and associated quartz porphyry dykes during a time when the field pointed towards the NE (Mertanen and Pesonen, 1995).

In our study of the Melville Bugt dykes, almost exactly the same observations hold: the dykes were intruded on either side of a reversal and the precise U-Pb data show convincingly that the NE-directed component is the younger, same as for the Sipoo dykes. Although the age data for the Sipoo dykes are less precise, they allow the same reversal to be recorded in the two studies. If so, then we have persuasive evidence that both Fennoscandia and Laurentia remained at 1.63 Ga in orientations not unlike their present ones and therefore that reconstructions that involve placing northern Norway and the Kola Peninsula adjacent to northern Greenland are incorrect. This interpretation can be tested by more precise age data and by more detailed paleomagnetic work on the Häme and Sipoo dykes.

The Melville Bugt and Sipoo swarms both have the same problem, namely that the NE-directed component may have a steeper negative inclination because of the difficulty in removing the strong PEF-like component. However the Melville Bugt dyke population shows several dykes (MB1, MB9, QT and JP) with steeper inclinations that are virtually devoid of PEF overprinting, suggesting that an underestimate of the inclination is minimal. However PEF contamination is more severe in the Fennoscandian dykes so that the shallow NE directions found in the Häme, Sipoo and Breven-Hällefors swarms, are more likely to be underestimating the inclination through failure to achieve a true end point. For the Sipoo dykes a further complication is the presence of a high coercivity remanence with a direction similar to that found as a secondary component at site MB3 (Table 27.1), so that in terms of coercivity the NE component is sandwiched between two remanences which have steep to moderate down inclinations with NE declinations.

The Fennoscandian paleomagnetic data closest in age to the Melville Bugt dykes are those from the 1,633 Ma Sipoo quartz porphyry and diabase dykes (Mertanen

and Pesonen, 1995). Specifically, we compare here the Sipoo pole position (26.4°N, 180.4°E) with that for Laurentia from the Melville Bugt dykes (pole 5°N, 273°E, Table 27.1) and also with that from the slightly younger 1,592 Ma Western Channel diabase (pole 6.8° N, 245°E, Irving et al., 1972; Hamilton and Buchan, 2010). The craton positions are shown in Fig. 27.9a–c where Fennoscandia, based on the Sipoo dykes data, occupies more equatorial positions than the two positions of Laurentia and therefore offers little support for a connection between the Melville Bugt dykes and the rapakivi province in Fennoscandia.

However in Fig. 27.9d and e we show a comparison that is closer. The position of Laurentia is adjusted using a pole (2°S, 267°E) that lies within the A_{95} of the Melville Bugt pole position in Table 27.1 (Fig. 27.9d). The streaking of site directions for Sipoo quartz porphyry dykes has a declination range of about 10° and an inclination range of ~30° from about +16 to –16° (Mertanen and Pesonen, 1995), inviting speculation that the true NE component is steeper. In fact one specimen upon thermal demagnetization (Fig. 27.5a of Mertanen and Pesonen, 1995) reaches an inclination of about –30°. If we assume a direction of 22°/–30° as closer to the true component direction, then a new pole position (11.5°N, 183.5°E) places Fennoscandia as shown in Fig. 27.9e. A further 10° increase in inclination to –40° (a few specimens attain these values, Satu Mertanen, personal communication 2010) would place the rapakivi province more or less in line with the Melville Bugt dykes of Fig. 27.9d. Although the correlation is far from perfect, a supportive observation is that the Western Channel Diabase data place Laurentia in a position such that the dykes trend more directly towards the Fennoscandian rapakivi province (c.f. Fig. 27.9a, e). The key element in our reconstruction is the assumption that the NE younger component in the Sipoo dykes has a steeper inclination. However if the rapakivi province extends beyond Fennoscandia with a southerly extension into Amazonia as shown in Fig. 27.1 of Johansson (2009) and the Melville Bugt dykes are fed from there, the positions of Laurentia and Fennoscandia as shown in Fig. 27.9b, c might require little adjustment.

Although the southern end of the Melville Bugt swarm disappears under ice, in SE Greenland high resolution satellite maps, courtesy Google Earth© (Figs. 27.10 and 27.11), image a NNW-trending series of dykes of unknown age, with widths up to 50 m, cutting basement gneisses, and approximately on strike with the Melville Bugt swarm. The dykes have a slightly reddish tinge with respect to the basement, with dyke-normal jointing suggesting that the dykes, like those of the Melville Bugt swarm, have not been metamorphosed. An investigation is needed to verify this correlation, but if true, it suggests that the Melville Bugt swarm could be continuous across southern Greenland and therefore allows it to be fed from the south.

That dykes can propagate laterally for over 2,000 km has been recognized by Ernst and Baragar (1992) from magnetic fabric studies on the giant 1.27 Ga Mackenzie swarm in Canada. Geochemical studies of Mesozoic Farrar diabase sills also suggest a lateral flow of about 2,500 km across Antarctica (Leat et al. 2006), so that a comparable distance of propagation that we advocate for the Melville Bugt dykes is not unreasonable.

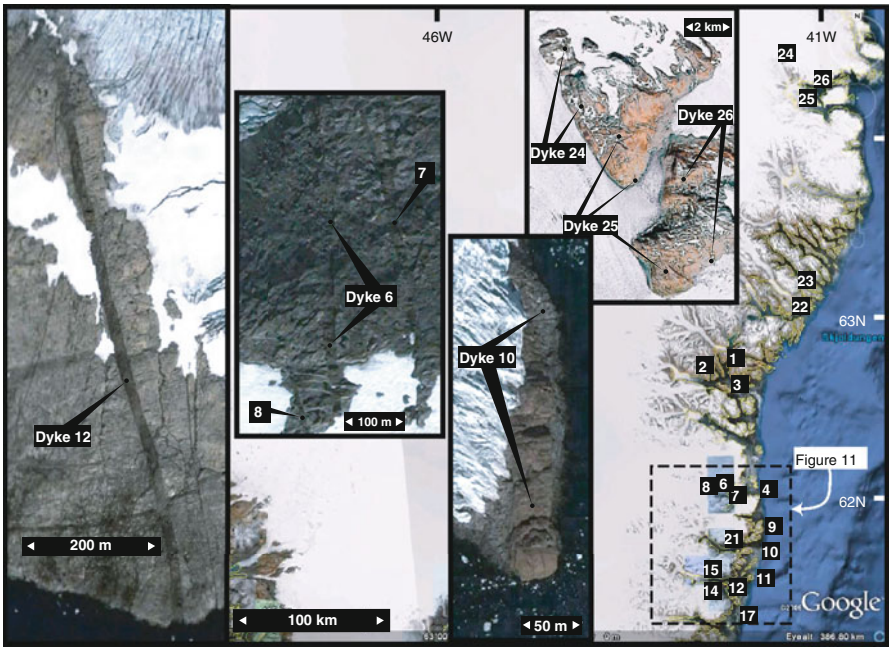


Fig. 27.10 Google Earth© images of SE Greenland showing evidence of NNW-trending dykes that may represent the southerly continuation of the Melville Bugt swarm. Most data come from relatively small survey areas of higher resolution (shown in *blue*). Overall coverage has relatively low resolution but in a few cases (sites 24 and 25) NW-trending dykes are visible (see farthest *inset* picture on the *right*). Lineaments with similar orientations occur at sites 22, 23 and 1–3, but their interpretation as dykes is less secure. However if the dykes exist a dyke swarm trending N to NW is more than 200 km wide, comparable to the minimum estimate for the Melville Bugt swarm

The geochemical study of the Melville Bugt master dyke (Kalsbeek and Taylor, 1986) shows that composition remains constant for 100's of km along strike and therefore that it may be similar to mafic dykes of the Fennoscandian rapakivi province, if we are correct in surmising that the two are genetically linked. We rely for geochemical estimates across the swarm on the work by Nielsen (1990), Denyszyn (2008), and on our unpublished major and trace element data from chilled margins of 10 Melville Bugt dykes. Melville Bugt dykes have P_2O_5 and TiO_2 compositions that average 0.50 ± 0.1 and 2.13 ± 0.2 wt% respectively (H.C. Halls, unpublished data from sites 1–9 and OF). Zr/Y values average 4.8 ± 0.5. One unusual aspect is that Ba concentrations average 870 ± 130 ppm. In Fennoscandia comparatively little work has been done on rapakivi-related diabase. For two Häme dykes P_2O_5 and TiO_2 average 0.50 and 2.3 wt% respectively and Zr/Y averages 7.9; Ba contents average about 500 (Rämö, 1990). However for the nearby (~100 km away) Suomenniemi swarm, also proximal to the Wiborg granite, analyses on five diabase dykes give the following: $P_2O_5 = 0.56$, $TiO_2 = 2.6$, $Zr/Y = 4.0$ and Ba averages

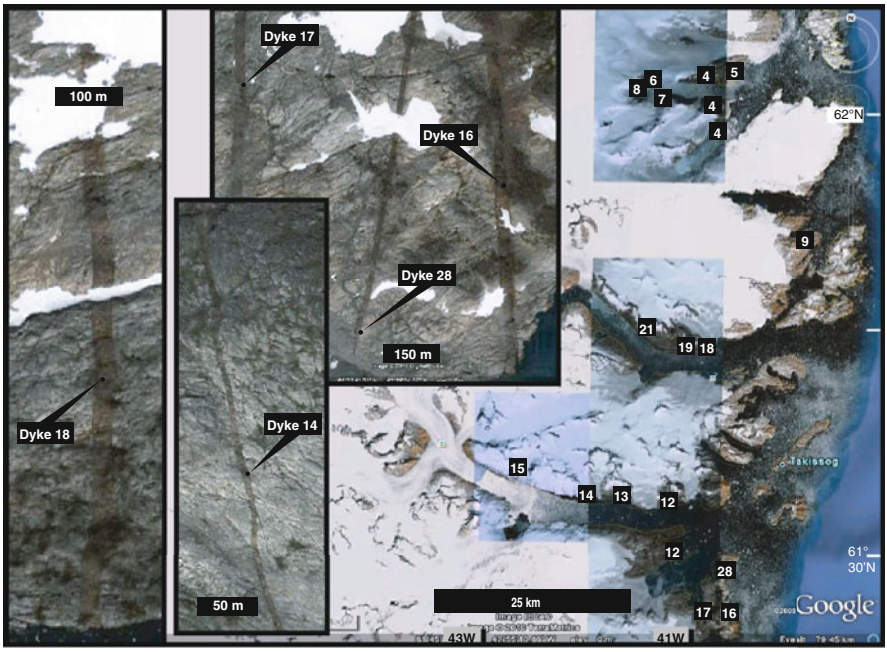


Fig. 27.11 Google Earth© images focusing more on the southern part of the dyke distribution shown in Fig. 27.10 Images are taken only from the higher resolution survey areas shown in *blue*

583 ppm with a maximum value of 809 ppm (Rämö, 1990). For the four major dykes of the Breven-Hällefors swarm, geochemical data reported by Risku-Norja (1992) give the following averages: $P_2O_5 = 0.36$, $TiO_2 = 2.1$ and $Zr/Y = 3.55$, with the Ba content remaining unmeasured. Therefore geochemically the dykes are surprisingly similar, being basalts that are all relatively rich in Ti and P and with Zr/Y ratios that mostly lie between 4 ± 1. However comparisons are flawed owing to the lack of correspondence between analysed and unmeasured elements, to inadequate sampling, and to differences in sampling philosophy (chilled margins in our study versus dyke interiors in the that by Risku-Norja).

The proposed reconstruction does not violate previous trans-Atlantic correlations of basement rocks older than 1.6 Ga (e.g. Zhao et al., 2004). In their reconstruction of the supercontinent Columbia Zhao et al. (their Fig. 17) show collisional orogens at 2.1–1.8 Ga and at 1.8–1.3 Ga which they correlate between Fennoscandia and Laurentia; both of these remain easily correlated in the configuration shown in Fig. 27.9. The configuration shown in Fig. 27.9 merely serves to reconfigure the Archean-Proterozoic margins and the 1.7 Ga Gothian margin with older rocks. In fact our reconstruction also resembles that proposed in Fig. 2 of Johansson (2009) for the time period 0.9 Ga, whereby the Gothian rocks exposed in Laurentia and Fennoscandia represent the *opposing* sides of the orogen. One puzzle is that during the lifetime of rapakivi magmatism (about 150 Myr), subduction-related accretionary magmatism was also taking place along the west coast of Norway

representing the later stages of the Gothian orogen (Åhäll et al., 2000) and signifying an ocean farther to the west. If, to the north, Greenland was separated from Fennoscandia by this ocean then a physical connection of the Melville Bugt swarm with the rapakivi igneous province would be invalidated. At the present time evidence for such a northward extension is lacking, thus allowing for dyke injection into Greenland from Fennoscandia.

Conclusions

Paleomagnetic and U-Pb geochronology demonstrate the sense of a magnetic polarity reversal found in the Melville Bugt dyke swarm and precisely date magmatic activity to span a time period of at least 13 million years as determined by maximum and minimum dates of 1,622 ± 3 and 1,635 ± 3 Ma. The paleomagnetic data strongly resemble those reported for the Sipoo dykes in Finland with an estimated U-Pb age of about 1,633 Ma (Mertanen and Pesonen, 1995), without any need for significant rotation of Laurentia; a NNE-N directed component with negative inclination and an older antipodal one are present in both data sets. If the same reversal is being recorded, it implies that Fennoscandia and Laurentia can be juxtaposed such that the southern end of the Melville Bugt dyke swarm can be projected to the proximity of the Fennoscandian rapakivi province. Paleomagnetic data can only be made to support this idea if magnetic overprinting of the Sipoo dykes has led to an underestimation of the negative inclination of the NE directed component. The correlation is best when Laurentia is positioned according to the 1,592 Ma Western Channel Diabase data rather than by those of the Melville Bugt dykes themselves. Diabase dykes associated with the rapakivi plutons have similar geochemical compositions to the Melville Bugt dykes. The enormous size of some of these dykes in Sweden indicates the capability of rapakivi magmatism to produce prodigious quantities of mafic magma and therefore our speculation that the entire Melville Bugt swarm is fed from a basaltic magma source below the continental crust that has ultimately produced the rapakivi granites, may be a reality, especially since the dykes may continue across Greenland. If so then the distance over which magma has been laterally injected, of the order of 2,000 km, offers support to other workers who also have proposed similar distances of basaltic magma transport. Finally our correlation of the Melville Bugt dykes with the Fennoscandian rapakivi granite province represents the first hypothesized Precambrian piercing point between Laurentia and Fennoscandia which is related to a specific magmatic event. The hypothesis is testable by further mapping in southeast Greenland and by more precise U-Pb dating, paleomagnetism and geochemistry of the Fennoscandian and Amazonian rapakivi units.

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